

GUI & Usability Testing for EShop Checkout

Human judgment, strengthened by AI - never replaced by it

Group 03 - CSC13003 - 23KTPM4

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Demo video: [VIDEO_LINK_REQUIRED](#)

A technically correct checkout can still fail its users

- Functional tests answer: “Did the action complete?”
- GUI testing asks: “Did the interface render and react correctly?”
- Usability testing asks: “Could a person understand and complete the task?”
- AI can propose issues quickly, but it cannot observe intent or own the final verdict.

Our approach combines real interaction, structured user research, and a human audit gate.

One checkout flow, three complementary lenses

Lens	Tool/practice	Main evidence
Cross-browser UI	BrowserStack Live	Reproduction on a named browser/device
Usability	Maze Free Explore	Click evidence + follow-up responses
AI-assisted review	Claude Vision	Screenshot-review hypotheses

Scope: cart -> login -> coupon -> amount review -> confirmation

Evidence becomes a defect only after human verification

1

Run EShop

2

Inspect in
BrowserStack

3

Capture evidence

4

Ask AI for hypotheses

5

Reproduce and
decide

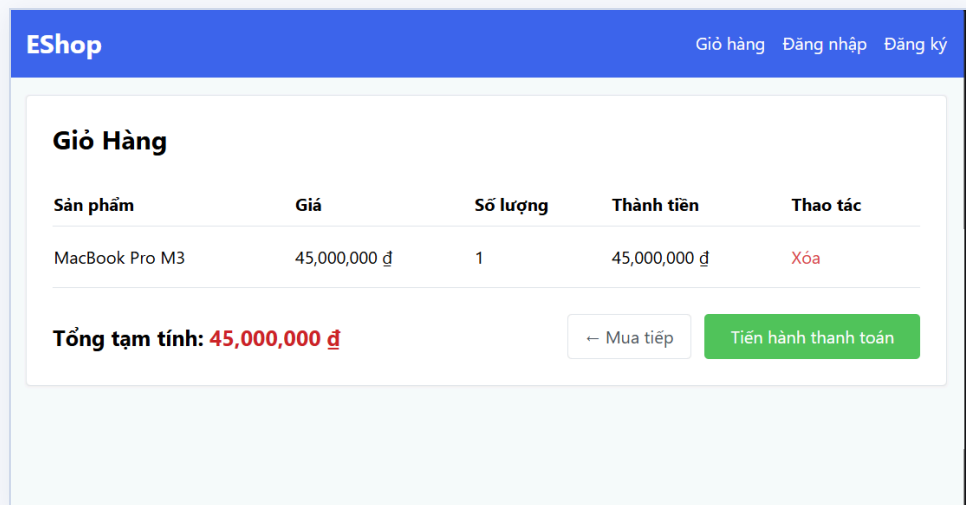
AI output is an input to testing, not a test result.

A small risk-based matrix is better than random coverage

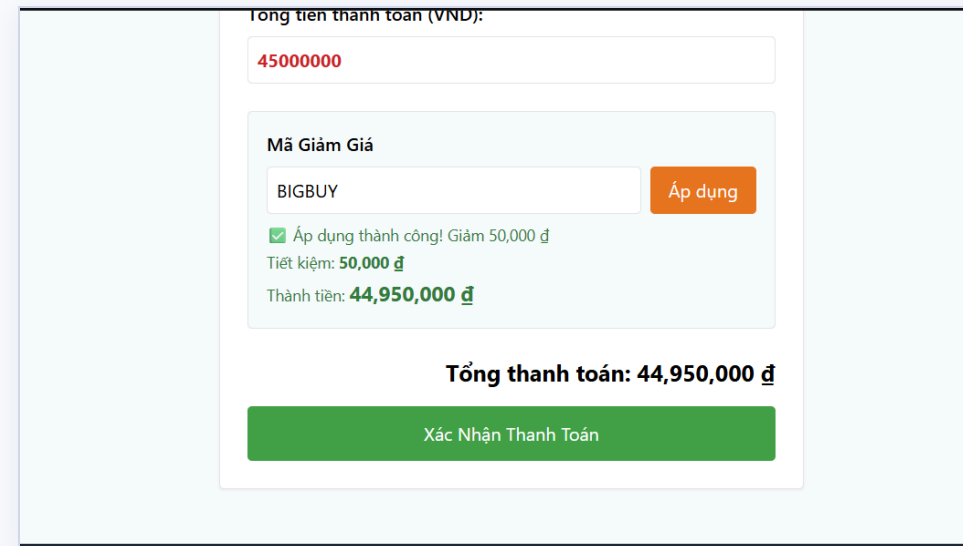
Environment	What it helps expose
Windows + Chrome	Primary desktop behavior
Windows + Firefox/Edge	Engine-specific CSS and focus behavior
macOS + Safari	WebKit rendering differences
Android device	Narrow viewport + virtual keyboard
iPhone + Mobile Safari	Touch targets + viewport behavior

Record exact OS, browser, version, device, viewport, account, and seed state for every finding.

The repository proves the checkout path was runnable



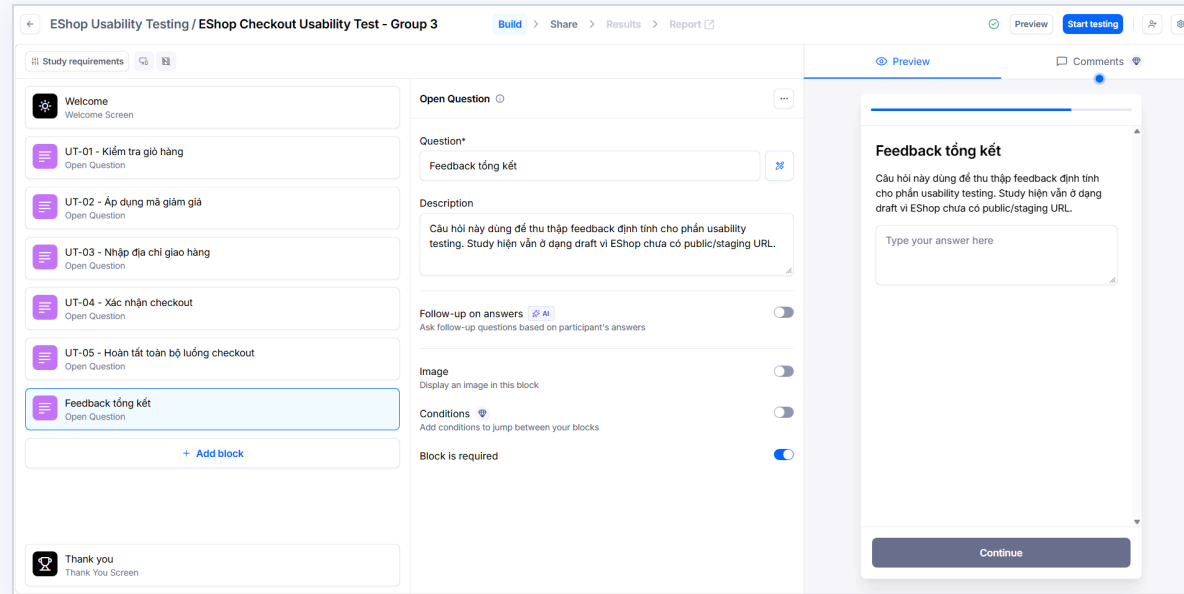
Local cart self-check



Local coupon self-check

These local screenshots demonstrate cart and coupon states. They are not cross-browser results and do not prove usability.

Maze turns the flow into an observable participant task



Maze study draft

Free Explore task

Add a product, reach checkout, apply the coupon, review the amount, confirm payment, and notice whether address confirmation exists.

Follow-up evidence

- Coupon-feedback clarity
- Address-step visibility
- Confidence after confirmation
- Overall ease
- Most effortful step
- One suggested improvement

Neutral wording protects the validity of the study

Avoid

Click “Giỏ hàng”, then press “Tiến hành thanh toán”, enter **BIGBUY**, and click the green button.

Prefer

Add one product, continue through checkout, apply the provided coupon, review the amount, and confirm payment.

Pilot with 1-2 people before release. Verify the URL, seed state, question order, and result capture.

Ask AI for hypotheses and the evidence needed

Prompt contract

Review this checkout screenshot as a hypothesis generator. For each possible issue, give the location, heuristic, tentative severity, evidence still needed, and a verification step. Do not claim a defect that cannot be established from the screenshot.

AI is useful for:

- Visual hierarchy and consistency checks
- Candidate heuristic violations
- Comparing desktop and mobile screenshots

AI cannot prove hover, focus, loading, DOM, keyboard, or user-intent claims from one screenshot.

The human audit gate has four possible verdicts

Verdict	Decision
Verified	Reproduced in the named environment
Rejected	Absent, expected, or based on a wrong assumption
Needs evidence	Requires interaction, DOM, or measurement
Duplicate	Same root cause as an existing issue

Final severity belongs to the human reviewer and is based on task impact, not AI confidence.

Each method has failure modes we must disclose

Method	Typical failure	Control
BrowserStack	Tunnel or shared-account state changes	Preflight and reset test data
Maze	Heatmap is mistaken for user intent	Triangulate with questions
AI vision	Hidden interaction problems are invented	Reproduce or reject
AI vision	Contrast ratio is asserted without measurement	Measure with a checker
Study design	Task wording teaches the solution	Pilot neutral wording
Sampling	A class sample is generalized	Label conclusions exploratory

Heuristic Showdown makes the audit process tangible

20-minute audience activity

1. Inspect one EShop screen and list at least five candidate issues.
2. Map each to a Nielsen heuristic and tentative severity.
3. Audit a prepared AI review containing plausible and unsupported claims.
4. Mark each claim: verified, rejected, or needs evidence.
5. Share one case where evidence changed the verdict.

The worksheet and answer key work offline after distribution.

Our strongest finding is about evidence discipline

The repository currently supports

- A runnable local checkout flow
- Cart, coupon, and confirmation screenshots
- A Maze study draft
- Documented AI-use evidence

It does not yet support

- Participant success or misclick rates
- A verified cross-browser defect set
- Claude precision/recall
- Fabricated audience feedback

Knowing what the evidence cannot say is part of good testing.

Use AI to widen the search - keep humans accountable

Observe with real interaction.

Ask AI for structured hypotheses.

Verify every claim.

Report limitations honestly.

Demo video (YouTube Unlisted): [VIDEO_LINK_REQUIRED](#)